

HDOC 24

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with options 1 to 6 to the user.
3. Read the user's choice.
4. Use a switch statement to call the respective function or exit if choice is 6.
5. Function **increasingStar()**:
 - Read n. Loop i from 1 to n. Inside, loop j from 1 to i. Print * . After inner loop, print newline.
6. Function **increasingRowNumber()**:
 - Read n. Loop i from 1 to n. Inside, loop j from 1 to i. Print i . After inner loop, print newline.
7. Function **increasingNaturalNumber()**:
 - Read n. Loop i from 1 to n. Inside, loop j from 1 to i. Print j . After inner loop, print newline.
8. Function **consecutiveNumber()**:
 - Read n. Initialize count = 1. Loop i from 1 to n. Inside, loop j from 1 to i. Print count++ . After inner loop, print newline.
9. Function **increasingAlphabet()**:
 - Read n. Validate $1 \leq n \leq 26$. Loop i from 1 to n. Inside, loop j from 1 to i. Print (char)('A' + j - 1). After inner loop, print newline.
10. Repeat steps 2-4 until choice == 6.
11. Stop

2. Test-Case Table

TC ID	Scenario Description	Inputs	Expected Output
TC-01	Right-angled triangle (Valid)	choice: 1, n: 4	4 rows of increasing asterisks
TC-02	Row-number triangle (Valid)	choice: 2, n: 4	Row 1: 1, Row 2: 2 2, Row 3: 3 3 3, Row 4: 4 4 4 4
TC-03	Natural-number pattern (Valid)	choice: 3, n: 4	Row 1: 1, Row 2: 1 2, Row 3: 1 2 3, Row 4: 1 2 3 4
TC-04	Consecutive-number triangle	choice: 4, n: 4	Row 1: 1, Row 2: 2 3, Row 3: 4 5 6, Row 4: 7 8 9 10
TC-05	Alphabet pattern (Valid)	choice: 5, n: 4	Row 1: A, Row 2: A B, Row 3: A B C, Row 4: A B C D
TC-06	Alphabet pattern (Boundary)	choice: 5, n: 30	Error message: n must be between 1 and 26.
TC-07	Invalid menu choice	choice: 8	Invalid choice. Please select 1-6.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes void
increasingStar(void);
void increasingRowNumber(void); void
increasingNaturalNumber(void); void
consecutiveNumber(void);
void increasingAlphabet(void);

int main() {
    int choice; do {
        printf("\n--- Increasing Patterns Menu ---\n"); printf("1. Increasing right-
        angled triangle\n"); printf("2. Increasing row-number triangle\n"); printf("3.
        Increasing natural-number pattern\n"); printf("4. Consecutive-number
        triangle\n"); printf("5. Increasing alphabet pattern\n"); printf("6. Exit\n");
        printf("Enter your choice: "); scanf("%d",
        &choice);

        switch (choice) {
            case 1: increasingStar(); break;
            case 2: increasingRowNumber(); break; case 3:
            increasingNaturalNumber(); break; case 4:
            consecutiveNumber(); break;
            case 5: increasingAlphabet(); break;
            case 6: printf("Exiting program...\n"); break;
            default: printf("Error: Invalid choice. Please enter 1-6.\n");
        }
    } while (choice != 6);

    return 0;
}

void increasingStar(void) { int n, i, j;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; } for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) { printf("* ");
        }
        printf("\n");
    }
}

void increasingRowNumber(void) { int n, i, j;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }
```

```

        for (i = 1; i <= n; i++) {
            for (j = 1; j <= i; j++) { printf("%d ", i);
                }
            printf("\n");
        }
    }
}

```

```

void increasingNaturalNumber(void) { int n, i, j;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; } for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) { printf("%d ", j);
            }
        printf("\n");
    }
}

```

```

void consecutiveNumber(void) { int n, i, j, count
    = 1;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; } for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) { printf("%d ", count++);
            }
        printf("\n");
    }
}

```

```

void increasingAlphabet(void) { int n, i, j;
    printf("Enter number of rows n (1 to 26): "); scanf("%d", &n);
    if (n < 1 || n > 26) {
        printf("Error: n must be between 1 and 26 for alphabet pattern.\n"); return;
    }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) { printf("%c ", 'A' + j - 1);
            }
        printf("\n");
    }
}

```